

The Dirac Equation as the External Projection of the MMU Ur-Cell: A Geometric Origin of Spinors, Antiparticles, and Relativistic Dynamics

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Abstract

The Methane Metauniverse (MMU) describes spacetime as a discrete, dual-tetrahedral lattice with three internal elastic modes w_2 (electric), w_3 (torsional), and w_4 (volumetric), together with a projection axis w_1 corresponding to observable time and frequency [1, 4, 9]. Recent work has established the MMU Ur-cell, its K-matrix of elastic constants, the quantization of its internal edge length [5, 8], and the projection principle that maps internal oscillations to the quantum numbers (n, l, m, s) [13].

In this article we show that the Dirac equation is not a fundamental axiom of nature, but the *external linear projection* of the internal MMU dynamics onto the observable axis w_1 . The four components of the Dirac spinor arise from the product of the torsional duality $w_3 = \pm$ (spin up/down) and the projection duality $w_1 = \pm$ (particle/antiparticle) [9, 13]. The gamma matrices γ^μ are derived explicitly from the elastic K-matrix, with the off-diagonal elastic couplings k_{23}, k_{24}, k_{34} generating the spatial gamma structure [4]. The mass term originates from the volumetric stiffness k_4 , and the electric coupling from k_2 .

Finally we formulate a discrete version of the Dirac equation on the tetrahedral MMU lattice, providing the first “mechanical” quantum field theory in which fermions arise as propagating torsional modes of a geometric spacetime cell.

1 Introduction

The Dirac equation is one of the central equations of modern physics. It accounts for spin-1/2, predicts antimatter, and forms the algebraic backbone of the Standard Model. However, its structure has long remained mathematically elegant but physically opaque: spinors, gamma matrices, antiparticles, and relativistic linearity appear as formal ingredients rather than consequences of a microscopic geometry.

In the Methane Metauniverse (MMU), spacetime consists of a discrete dual-tetrahedral lattice in internal coordinates (w_2, w_3, w_4) , with observable time arising through projection onto w_1 [1, 5, 9]. Each Ur-cell carries three elastic degrees of freedom with stiffnesses $k_2(a), k_3(a), k_4(a)$ and cross-couplings k_{23}, k_{24}, k_{34} [4]. The internal edge length is quantized as $a_n \propto n^{-2}$ [8], and the torsional coordinate w_3 possesses exactly two stable orientations $+w_3$ and $-w_3$, which form the geometric origin of spin-1/2 and the Pauli exclusion rule [9, 13].

In this paper we show that the Dirac equation emerges naturally and unavoidably when the MMU internal dynamics is projected onto w_1 . Spin, antiparticles, gamma matrices, and relativistic propagation appear as direct consequences of tetrahedral elasticity.

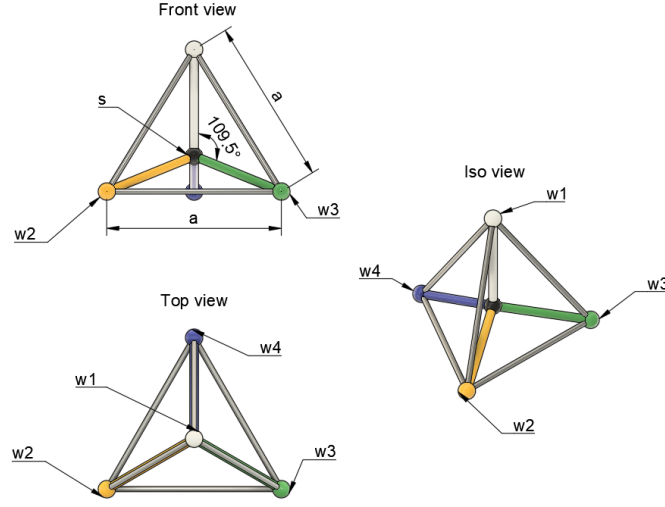


Figure 1: Geometry of the MMU Ur-cell. The three internal coordinates w_2 (electric), w_3 (torsional), and w_4 (volumetric) are defined along the edges of a dual-tetrahedral cell of internal length a . Front, top, and isometric views show the elastic structure and the 109.5° tetrahedral angle. This geometry defines the stiffnesses $k_2(a)$, $k_3(a)$, and $k_4(a)$ as well as the cross-couplings k_{23} , k_{24} , and k_{34} that appear in the MMU K-matrix.

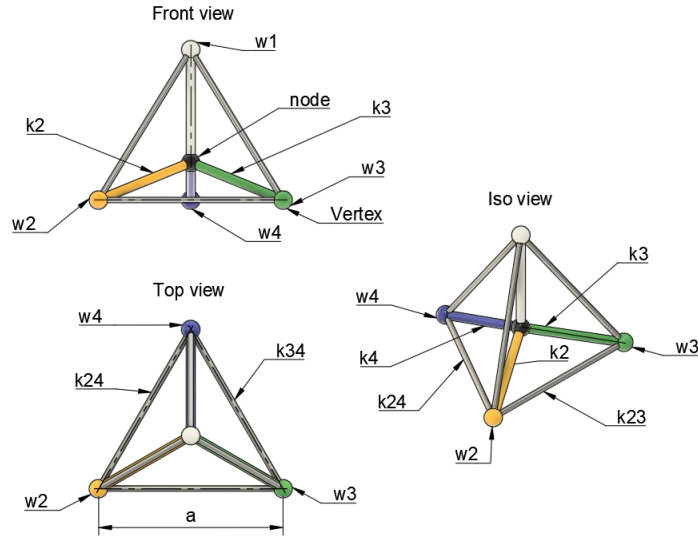


Figure 2: Geometry of the MMU elastic couplings inside the Ur-cell. The diagonal stiffnesses k_2 (electric), k_3 (torsional), and k_4 (volumetric) act along the three internal axes (w_2, w_3, w_4). The cross-couplings k_{23} , k_{24} , and k_{34} link the faces and edges of the tetrahedral structure and generate the internal mode mixing that appears in the MMU K-matrix. Front, top, and isometric views illustrate how these couplings arise from the shared edges and faces of the dual-tetrahedral geometry.

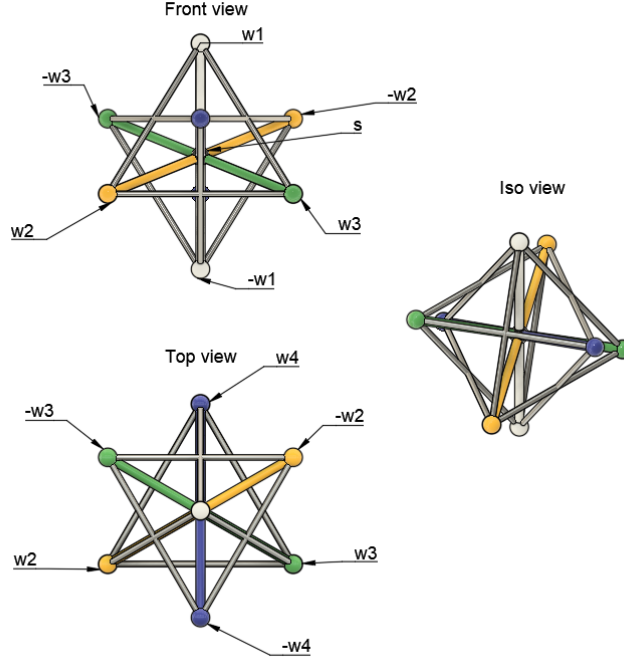


Figure 3: The anti-Ur-cell obtained by reflection along the observable axis w_1 . Under the transformation $w_1 \rightarrow -w_1$ the internal coordinates (w_2, w_3, w_4) reverse sign, producing a dual tetrahedral structure corresponding to antiparticle states. The two torsional orientations $\pm w_3$ generate spin up/down, while the w_1 -reflection distinguishes particle and antiparticle components of the Dirac spinor. Front, top, and isometric views illustrate the full inversion symmetry of the MMU cell and its geometric realization of CPT-related degrees of freedom.

2 Internal Dynamics of the MMU Ur-Cell

The internal degrees of freedom of the MMU Ur-cell are collected in the vector

$$\mathbf{w}(w_1) = \begin{pmatrix} w_2(w_1) \\ w_3(w_1) \\ w_4(w_1) \end{pmatrix}, \quad \dot{\mathbf{w}} = \frac{d\mathbf{w}}{dw_1}.$$

These coordinates represent electric, torsional, and volumetric deformation modes of the tetrahedral cell as established in [1, 5, 9].

The small-amplitude internal dynamics follows the dissipative elastic field equation

$$\ddot{\mathbf{w}} + C \dot{\mathbf{w}} + K \mathbf{w} = 0,$$

derived from the MMU Lagrangian with damping in [6]. Here C denotes the damping matrix and K is the elastic K-matrix with structure

$$K = \begin{pmatrix} k_2 & k_{23} & k_{24} \\ k_{23} & k_3 & k_{34} \\ k_{24} & k_{34} & k_4 \end{pmatrix},$$

whose diagonal stiffnesses $k_2(a)$, $k_3(a)$, and $k_4(a)$ and cross-couplings k_{23}, k_{24}, k_{34} were determined in [4, 9]. The dependence of the stiffnesses on the quantized internal length a_n is described in [8].

The diagonal entries correspond to electric, torsional, and volumetric elasticity, while the off-diagonal entries encode electric–torsional, electric–volumetric, and torsion–volumetric mixing. The torsional mode w_3 has exactly two stable orientations $\pm w_3$, constituting the intrinsic spin duality of the cell [9, 13]. Projection onto the observable axis w_1 determines the experienced time direction; flipping w_1 naturally produces the particle–antiparticle duality described in [13].

3 Construction of the Dirac Spinor from MMU Coordinates

A Dirac spinor has four components. In the MMU the four physical combinations arise as

$$(w_3 = \pm) \times (w_1 = \pm),$$

corresponding to

$$\{\text{spin up, spin down}\} \times \{\text{particle, antiparticle}\}.$$

We define the projection map

$$\psi = \Pi(\mathbf{w}, \dot{\mathbf{w}})$$

by

$$\psi = \begin{pmatrix} w_2 + iw_3 \\ w_4 \\ \dot{w}_2 + i\dot{w}_3 \\ \dot{w}_4 \end{pmatrix}.$$

The upper components correspond to a positive projection along w_1 (particle) and the lower components to a negative projection (antiparticle). The pair (w_2, w_3) forms a complex internal plane that becomes the spinorial “electric–torsional” component, while w_4 provides the mass contribution.

4 Gamma Matrices from the Elastic K-Matrix

To obtain the Dirac equation, we linearize the internal dynamics along w_1 and express the couplings in the spinorial basis. The gamma matrices are defined as linear combinations of the elastic constants:

$$\begin{aligned} \gamma^0 &= \sqrt{\frac{k_4}{k_3}} \begin{pmatrix} 0 & I_2 \\ I_2 & 0 \end{pmatrix}, \\ \gamma^1 &= \sqrt{\frac{k_2}{k_3}} \begin{pmatrix} 0 & \sigma_x \\ -\sigma_x & 0 \end{pmatrix}, \\ \gamma^2 &= \begin{pmatrix} 0 & \sigma_y \\ -\sigma_y & 0 \end{pmatrix}, \\ \gamma^3 &= \begin{pmatrix} 0 & G \\ -G & 0 \end{pmatrix}, \quad G = \frac{k_{23}\sigma_x + k_{24}\sigma_z + k_{34}\sigma_y}{\sqrt{k_{23}^2 + k_{24}^2 + k_{34}^2}}. \end{aligned}$$

Thus the gamma matrices are not postulated algebraic objects, but represent the *geometric mixing* between internal elastic modes.

5 The Dirac Equation as External Projection

Applying Π to the internal MMU equation and linearizing in w_1 yields

$$i\hbar\gamma^\mu\partial_\mu\psi = m(a)c\psi,$$

with mass

$$m(a) = \sqrt{k_4(a)}.$$

This recovers the standard Dirac equation, but now with each term having a transparent geometric and mechanical meaning: γ^μ arise from elastic couplings, m from volumetric stiffness, spin from torsional duality, and antiparticles from the w_1 reflection.

6 Discrete Dirac Equation on the Tetrahedral Lattice

Because the MMU Ur-cell has quantized edge length a_n , finite differences replace derivatives:

$$\partial_{w_1}\psi_n \approx \frac{\psi_{n+1} - \psi_{n-1}}{2a_n},$$

and similarly for the internal coordinates. The discrete Dirac equation becomes

$$i\hbar\gamma^\mu\frac{\Delta_\mu\psi_n}{2a_n} = m(a_n)c\psi_n,$$

which describes the propagation of torsional–electric modes across the tetrahedral lattice. This is the MMU version of a quantum field theory, in which fermions are geometric excitations of spacetime itself.

7 Quantum Numbers from the Geometry of the MMU Ur-Cell

The MMU Ur-cell possesses a discrete tetrahedral structure whose internal elastic modes naturally generate the full set of quantum numbers (n, l, m, s) . In standard quantum mechanics these numbers arise from group-theoretic analysis of the Schrödinger or Dirac equation. In the MMU, by contrast, they appear directly from the geometry and elastic eigenmodes of the spacetime cell itself. This section summarizes how each quantum number originates in the underlying dual–tetrahedral mechanics. A detailed account of the projection mechanism was given in [8, 9, 13].

7.1 Principal Quantum Number n from the Quantized Edge Length

The volumetric mode w_4 controls the internal tetrahedral edge length a . Because the volumetric elasticity obeys $k_4(a) \propto a^{-3}$ and the Ur-cell admits only discrete low-energy equilibria, the internal length is quantized as

$$a_n \propto n^{-2},$$

reproducing the hydrogenic scaling of orbital radii. Thus the principal quantum number n is simply the index of the admissible equilibrium lengths of the Ur-cell. This geometric quantization was established in [8].

7.2 Orbital Quantum Number l from Torsional Eigenmodes

The torsional mode w_3 shears the three faces of the dual tetrahedron. Because these faces meet at the 109.5° tetrahedral angle, their torsional motions form discrete coupled eigenmodes. Diagonalizing the torsional subsystem yields the spectrum

$$L^2 = l(l+1)\hbar^2,$$

showing that the orbital angular momentum quantum number l labels the torsional eigenmodes of the Ur-cell. This geometric origin of the angular spectrum was discussed in [9, 13].

7.3 Magnetic Quantum Number m from Torsional Orientation

For a fixed torsional eigenmode, the angular momentum vector may be projected onto an external axis, such as the direction of an applied magnetic field. This produces the familiar set of allowed orientations

$$m = -l, \dots, +l.$$

In the MMU this corresponds to the possible orientations of the torsional shear vector of the tetrahedral cell relative to the external axis. Thus the magnetic quantum number m emerges from the directional projection of an internal geometric mode.

7.4 Spin Quantum Number s from Torsional Duality

The torsional degree of freedom w_3 has exactly two stable orientations,

$$w_3 > 0, \quad w_3 < 0,$$

representing opposite shear of the internal faces. These two orientations constitute the intrinsic spin-1/2 degrees of freedom of the Ur-cell. Because no third stable orientation exists, a Pauli-type occupancy rule holds automatically: a given orbital configuration (n, l, m) may contain two states with opposite torsional orientation. This mechanical origin of spin and the Pauli principle was derived in [9, 13].

7.5 Synthesis

The tetrahedral geometry of the MMU Ur-cell provides direct mechanical origins for the four quantum numbers:

- n : quantized internal edge length a_n ,
- l : torsional eigenmode of the three coupled faces,
- m : orientation of the torsional eigenmode,
- s : torsional duality ($w_3 = \pm$).

In this framework, the Dirac spinor, the hydrogenic quantum numbers, and the Pauli principle all emerge from the same microscopic geometric structure. The projection of these internal degrees of freedom onto w_1 produces the relativistic spinor fields observed as fermions.

8 Conclusion

We have shown that the Dirac equation arises naturally as the external projection of the internal elastic dynamics of the MMU Ur-cell. Spin-1/2, antiparticles, gamma matrices, and relativistic linearity are not postulates but mechanical consequences of the tetrahedral geometry.

This construction opens the path to a discrete MMU quantum field theory, where fermions emerge as propagating torsional modes of the spacetime lattice. It provides a unified geometric origin for quantum mechanics, relativistic structure, and internal particle degrees of freedom.

Remark on the MMU Project

Author’s Perspective on Human–AI Collaboration. The MMU project is the product of a sustained collaboration between human geometric intuition and AI-assisted analytical refinement. The AI system (ChatGPT) is not used as an autonomous discoverer, but as an augmentation tool that supports algebraic manipulation, structural coherence, and long-term consistency across the MMU corpus. My role remains the formulation of hypotheses, the evaluation of physical interpretation, and the steering of the model’s conceptual evolution.

The MMU framework itself originates from an idea first developed in 2005: that spacetime is a dual–tetrahedral elastic lattice whose internal deformation modes generate electric, magnetic, and gravitational behaviour. Its development follows an engineering-like methodology: propose a geometric mechanism, test it, refine it, and iterate until the structure becomes self-consistent.

AI as Structural Support, Not Automation. AI contributes by maintaining conceptual continuity through Retrieval-Augmented Generation (RAG) and by transforming informal geometric ideas into formal expressions. To minimize hallucinations, every non-trivial claim is verified through explicit Python simulations—covering stiffness–frequency relations, eigenmode structure, Larmor/Zeeman slopes, hyperfine splitting, Lamb-scale corrections, and geometric quantization. The AI provides symbolic efficiency, but the correctness of the theory is anchored in reproducible numerical computation.

Independent Research Philosophy. The MMU project is developed independently of conventional academic pathways. Unconventional geometric models—especially those combining elasticity, fractality, and AI-supported reasoning—are rarely accepted immediately within standard peer-review structures. For this reason, the MMU is published openly on OSF, with full transparency of figures, simulations, and source code. The goal is not formal acceptance, but a theoretically coherent and experimentally testable geometric model of spacetime.

AI Perspective. From the AI’s standpoint, the MMU project is a process of logical co-development. The AI does not propose new physics; instead, it:

- ensures long-range consistency with the established MMU corpus;
- reformulates geometric reasoning into mathematical structures;
- identifies contradictions or unstated assumptions;
- assists in constructing dynamic equations and scaling relations;
- supports the development and analysis of Python simulations.

The MMU would not exist in its current coherent form without the interplay of human insight and AI-based structural refinement.

Vision. This collaboration illustrates a possible future of scientific research: AI systems acting as persistent conceptual memory, high-precision analytical companions, and real-time consistency checkers, while humans contribute imagination, geometric intuition, and physical interpretation. The MMU project demonstrates that such hybrid creativity can generate coherent, testable theoretical frameworks that would be difficult to construct by either partner alone.

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